

Influence of atmospheric turbulence and vertical wind on noctilucent clouds

Gokul Udayakumar^a

^a*School of Integrated Climate System Sciences, , University of Hamburg, , ,*

Abstract

Noctilucent clouds (NLCs) near the polar summer mesopause are highly sensitive to changes in atmospheric dynamics, including turbulence and vertical wind. In this study, we use the Mesospheric Ice Microphysics And Transport (MIMAS) model to simulate how variations in turbulence and vertical wind conditions affect the formation and properties of NLCs. Specifically, we conducted multiple simulations with different vertical wind scenarios and turbulence profiles. By comparing the resulting cloud particle distributions, occurrence frequencies, and spatial patterns, we aim to understand the relative roles of these factors in shaping NLCs. Our approach seeks to enhance our understanding of how variations in vertical wind and turbulence influence NLC variability, particularly in relation to gravity wave dynamics.

Keywords: Turbulence, NLC, Vertical wind, MIMAS

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1. Introduction

Noctilucent clouds (NLCs), also referred to as polar mesospheric clouds (PMCs), are the highest known clouds in Earth’s atmosphere, forming near the mesopause region at approximately 80 to 86 km altitude Jesse (1885), Backhouse (1885)

,Leslie (1885) They consist of ice particles typically measuring 40 to 100 nm in diameter, significantly smaller than tropospheric cloud particles, which can exceed 10 μm in size. These nanoscale ice particles grow under extremely cold conditions ($< 150\text{ K}$), where even a few parts per million of water vapor can reach supersaturation Thomas (1996) Observed predominantly at high latitudes ($\gtrsim 55^\circ$) during the summer season. NLCs appear with a distinct silvery-blue glow under nautical twilight, when the lower atmosphere and ground are already in darkness but the upper atmosphere is still sunlit. Despite possibly occurring during daylight, they cannot be easily seen without specialized instruments because of strong Rayleigh scattering below the mesosphere.

Their discovery is historically linked to the eruption of Krakatoa in 1883, as increased volcanic aerosols may have contributed to early records of these phenomena in 1885 Leslie (1885),Backhouse (1885) Contemporary techniques for studying NLCs encompass an array of ground and space-based instrumentation. From the ground, lidars (LIght Detection And Ranging) can detect mesospheric ice both day and night by measuring backscattered laser light Fiedler et al. (2009) whereas conventional cameras are limited to twilight observations. Satellite missions such as the Aeronomy of Ice in the Mesosphere (AIM) capture images in the ultraviolet spectrum (e.g., the Cloud Imaging and Particle Size experiment, CIPS) or measure solar occultation (SOFIE) to infer ice particle extinction properties McClintock et al. (2009),Hervig et al. (2009) Russell III et al. (2009). Additionally, radar-based methods detect small ice particles through their influence on polar mesospheric summer echoes (PMSE), which frequently coincide with NLC altitudes Nussbaumer et al. (1996), Gumbel and Witt (2001), Rapp and Lübken (2004) Kaifler et al. (2011).

Figure 1 in the referenced material summarizes the microphysical and optical processes that govern NLC. Ice particles nucleate around meteoric smoke particles in the extremely cold

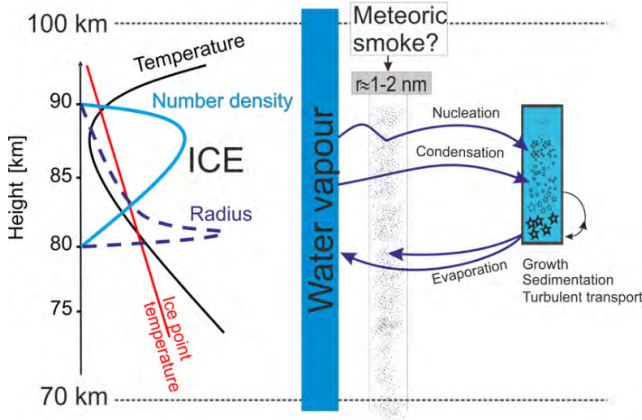


Figure 1: Summary of microphysical processes involved in the formation of noctilucent clouds, adapted from Baumgarten (2001) and Rapp and Thomas (2006).

and dry environment of the mesopause, where temperatures can drop to 128 to 150 K Lübken (1999). Although such particles initially remain small (< 10 nm) due to the paucity of water vapor at higher altitudes, some become transported down to approximately 82 to 83 km by sedimentation, vertical winds, and turbulent mixing. In this slightly warmer but still supersaturated region, particles can grow to visible sizes (around 40 to 100 nm), efficiently scatter sunlight in the Mie regime, and form the noctilucent clouds we observe Hervig et al. (2009), Baumgarten et al. (2012). As these particles descend further into undersaturated air, sublimation releases water vapor back into the atmosphere. This feedback cycle illustrates how NLC evolution depends sensitively on microphysical processes and dynamical drivers such as vertical winds and turbulence.

Recent decades have seen increasing NLC occurrences at slightly lower latitudes and with higher frequency, fueling speculation that these clouds may serve as an early indicator of climate change in the upper atmosphere Thomas (1996), Lübken et al. (2018). Indeed, their pronounced sensitivity to temperature and water vapor has long established NLCs as natural tracers of mesospheric conditions Hervig et al. (2009). However, while temperature and water vapor remain key controlling variables, mounting evidence points to critical roles for turbulence and vertical wind in shaping ice particle nucleation, growth, and transport Kiliani et al. (2013). In particular, gravity waves, generated by processes in the lower atmosphere, propagate upward and break in the mesosphere, depositing energy and momentum Thayer et al. (2003), Baumgarten and Fritts (2014), Klostermeyer (1998). This breaking drives both large-scale circulations and localized turbulence, enhancing vertical mixing and potentially lifting particles into colder regions or sinking them into warmer ones, thus modulating the visibility and lifetime of NLCs Jensen (1989).

In this study, we employ the three-dimensional Mesospheric Ice Microphysics And tranSPort (MIMAS) model Berger and Von Zahn (2002) to explore how turbulence and vertical wind variations affect NLC formation and distribution. By isolating these individual drivers, we aim to provide new insights into

the interplay among gravity waves, turbulent mixing, and vertical motion in shaping the morphology and longevity of mesospheric ice clouds. Beyond advancing our understanding of NLC microphysics, this work also addresses broader questions regarding mesospheric response to a changing climate. Improved knowledge of these processes may ultimately enhance the representation of upper-atmospheric phenomena in global climate models, helping us better predict and interpret the evolution of our planet's climate system.

2. Study Objectives

2.1. Objective 1: Influence of Turbulence on NLC Formation and Evolution

The first objective of this study is to determine how varying degrees of atmospheric turbulence affect the formation, evolution, and overall characteristics of noctilucent clouds (NLCs). Turbulence in the mesosphere is often generated by breaking gravity waves, which deposit momentum and energy at high altitudes, leading to enhanced mixing of water vapor and ice particles. Observational evidence from high-resolution lidar measurements illustrates this process vividly; for instance, **Figure 2** shows an NLC with wave-like overturning structures (indicated by the arrow), reminiscent of breaking waves on a beach. Such structures testify to the turbulent cascades that can disrupt stable stratification, allowing supersaturated regions to expand vertically and horizontally Kiliani et al. (2013). This enhanced mixing has significant implications for the microphysical processes responsible for NLC formation: as water vapor is redistributed, supersaturation may persist or even intensify in regions that would otherwise be too warm or too dry for cloud development. Meanwhile, turbulence-driven diffusion can carry ice particles downward into warmer layers, accelerating sublimation and reducing their overall lifetime Rapp and Lübken (2004), Rapp et al. (2021). These competing effects enhance nucleation versus faster sedimentation and sublimation highlight the complexity of turbulence–cloud interactions in the upper atmosphere.

By incorporating a range of turbulence intensities into the MIMAS model, we aim to capture the full spectrum of possible outcomes on NLC brightness, particle size distribution, and geographic extent. Model simulations under progressively stronger or weaker turbulent conditions will allow us to identify thresholds where increases in turbulence either boost or diminish NLC visibility. Such sensitivity studies are crucial for evaluating how existing parameterizations of turbulence influence mesospheric cloud simulations in global climate models, which often lack the fine-scale resolution needed to explicitly resolve wave-breaking phenomena Fritts and Alexander (2003). By comparing these simulated NLC responses to observational data (e.g., lidar backscatter profiles, satellite UV scattering measurements), we can determine the level of turbulence detail required to reproduce key features of cloud morphology and seasonal variability. Ultimately, this objective provides new insights into the coupling between small-scale turbulence and the larger-scale dynamical and thermodynamical state of the

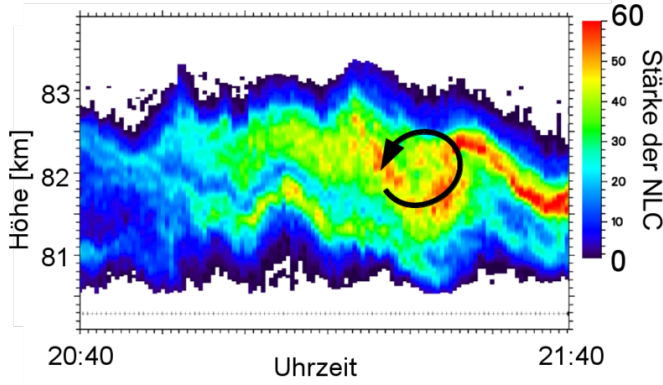


Figure 2: Observation of a noctilucent cloud by lidar. The arrow indicates overturning structures similar to breaking waves at the beach. Such evidence of wave-breaking highlights the importance of turbulent processes in NLC formation and evolution.(the above image is obtained from IAP)

mesosphere, thereby refining our ability to predict and interpret NLC behaviors in a changing climate.

2.2. Objective 2: Role of Vertical Winds in NLC Formation and Persistence

A second objective of this study is to evaluate how vertical winds influence the formation, maintenance, and dissipation of noctilucent clouds (NLCs). Vertical wind perturbations in the upper atmosphere are often driven by gravity waves, which can propagate upward from tropospheric or stratospheric sources and deposit energy as they break Thayer et al. (2003), Baumgarten and Fritts (2014), Fritts and Alexander (2003). Even relatively modest upward velocities (on the order of a few centimeters per second) can lift ice particles and water vapor to colder altitudes, extending the time spent in supersaturated conditions and thus facilitating particle growth to visible sizes Jensen (1989). Conversely, downward winds can quickly transport particles into warmer regions, enhancing sublimation and diminishing NLC brightness or even causing the clouds to dissipate Kirkwood et al. (2015)

By systematically varying the vertical wind parameter in the MIMAS model, we aim to quantify the extent to which different upward and downward motions modulate the microphysical processes governing ice particle lifecycles. Such controlled simulations will help isolate how changes in vertical velocity influence nucleation, growth, and sedimentation rates. Moreover, comparing model outputs to observational constraints—such as lidar backscatter ratios or satellite observations of cloud brightness—will offer insights into the adequacy of current vertical wind representations in mesospheric simulations Berger and Von Zahn (2002), Kiliani et al. (2013). Ultimately, improved understanding of the interplay between vertical wind patterns and NLC microphysics will enable a more accurate depiction of mesospheric dynamics in climate models and shed light on how changing atmospheric conditions may alter NLC frequency and visibility in the future.

2.3. Objective 3: Validating the model with Observations

A further objective of this study is to assess the fidelity of the Mesospheric Ice Microphysics And tranSPort (MIMAS) model in reproducing real noctilucent cloud (NLC) properties by leveraging comprehensive day-and-night lidar and radar measurements obtained at the Leibniz Institute of Atmospheric Physics. These dual-mode observations provide continuous data on cloud brightness, altitude profiles, particle number density, and polar mesospheric summer echoes (PMSE) associated with NLCs Fiedler et al. (2009), Kaifler et al. (2011). Such synergy of lidar and radar instrumentation enables both daytime and nighttime detection of mesospheric ice layers, ensuring that critical microphysical and dynamical details—such as the presence of turbulent eddies or local vertical wind fluctuations—are captured across varying temporal and lighting conditions.

In this context, comparing MIMAS-simulated NLC events against these institute-collected datasets allows for a detailed evaluation of the model’s performance. Discrepancies in temporal evolution, spatial distribution, or cloud intensity would point to needed refinements in the model’s turbulence and vertical wind parameterizations. Conversely, robust agreement would reinforce confidence in MIMAS as a predictive tool for probing the interplay among microphysics, dynamics, and climate-driven changes in the mesosphere. By integrating these high-resolution observations and model simulations, this objective ultimately seeks to advance both the reliability of MIMAS and the broader understanding of how NLCs respond to diverse atmospheric forcing mechanisms.

2.4. Additional Objective: Translating the MIMAS Code from Fortran 90 to Julia

Although not formally listed among the primary scientific objectives, this final component of the master’s thesis is of considerable importance for the broader applicability of the Mesospheric Ice Microphysics And tranSPort (MIMAS) model. Specifically, the goal is to translate the existing MIMAS Fortran 90 codebase into the Julia programming language. Julia has gained prominence in the scientific computing community due to its ease of use, speed, and ability to integrate seamlessly with other modern numerical libraries. By reimplementing MIMAS in Julia, the model stands to benefit from more transparent syntax, improved maintainability, and potentially higher performance through just-in-time (JIT) compilation. Furthermore, Julia’s growing ecosystem simplifies collaboration by enabling straightforward coupling with other transport or atmospheric models, facilitating multi-disciplinary research on a larger scale.

This conversion effort aims to preserve the physical integrity and accuracy of MIMAS’ microphysical routines, ensuring that its capacity to simulate noctilucent cloud formation remains intact. Beyond that, transitioning the model to a more accessible and modular framework will encourage a wider adoption within the scientific community—especially among researchers looking to integrate advanced ice microphysics into their mesospheric or global circulation models. Ultimately, this work

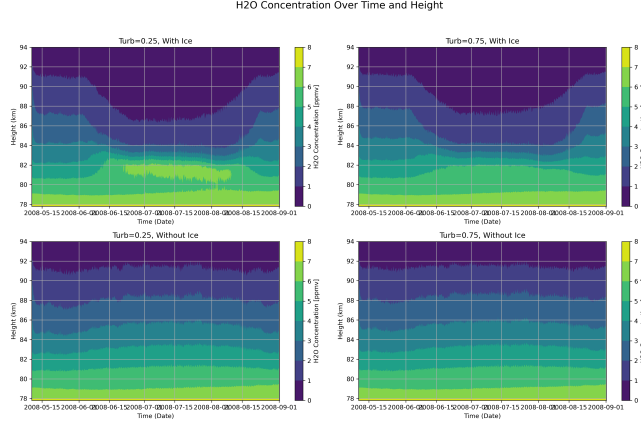


Figure 4: Time-height cross-sections of water concentration for turbulence intensities at 25% (left) and 75% (right) of the baseline.

4. Expected Results

Disclaimer Regarding Preliminary Results

The results and plots presented in this section originate from preliminary simulations conducted by Dr. Ashique Velalassery, utilizing variations of turbulence intensity at 25% and 75% of baseline levels. The data shown here have been plotted by the author solely for the purpose of verifying model outputs and becoming familiar with the simulation data structure generated by MIMAS. It must be emphasized that these simulations do not represent the final experimental configuration intended for this project. The formal experiments designed to systematically investigate the impact of turbulence and vertical winds on noctilucent clouds have not yet commenced. Accordingly, these preliminary findings will be revised and updated upon completion of the full set of planned model experiments.

4.1. Profiles of Water Vapor

Figure 5 presents the vertical profiles of H_2O concentration, averaged over latitude and longitude for the summer months (June, July, and August). The profiles are shown for two different turbulence cases: 25% and 75% of the baseline turbulence levels. The solid and dashed lines represent the cases with and without ice formation, respectively. A clear reduction in water vapor concentration is observed at higher altitudes when ice particles are present, indicating active condensation and subsequent growth of NLC ice particles. This reduction is more pronounced in the lower turbulence case, suggesting that weaker turbulence results in a more localized and stable ice formation region. In contrast, enhanced turbulence (75% case) leads to a slightly broader altitude range where water vapor is reduced, likely due to increased mixing and redistribution of H_2O in the mesosphere.

The time evolution of mesospheric water vapor is depicted in Figure 4, which shows time-height contour plots of H_2O concentration over the entire simulation period. These plots further illustrate the impact of ice formation on the vertical distribution of water vapor. The upper altitudes exhibit a noticeable

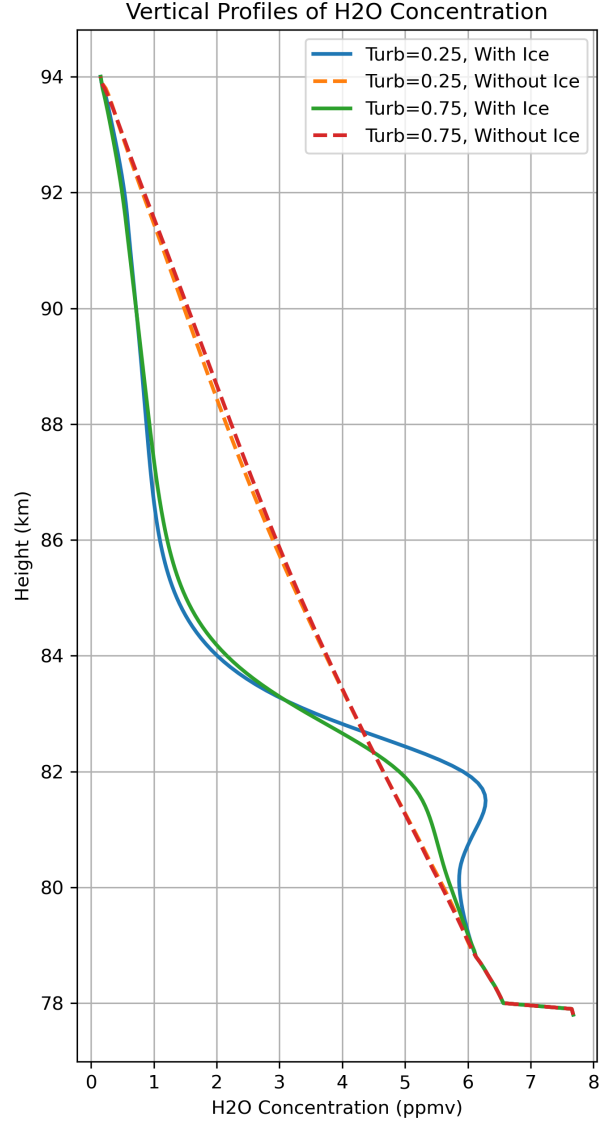


Figure 5: Vertically averaged profiles of water vapour concentration for the 25% (blue) and 75% (orange) turbulence cases.

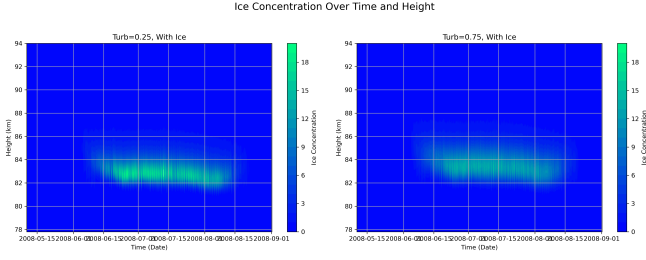


Figure 6: Time-height cross-sections of ice concentration for turbulence intensities at 25% (left) and 75% (right) of the baseline.

decrease in H_2O concentration, particularly in the presence of ice, supporting the idea that ice nucleation and growth actively deplete available water vapor at these heights. Additionally, the variations over time reflect seasonal and dynamical influences on water vapor transport, with increased turbulence leading to a more diffused distribution of moisture. The comparison between the turbulence cases emphasizes the role of mixing in shaping the spatial and temporal variability of water vapor in the mesosphere.

4.2. Formation and Vertical Distribution of ice

Figure 6 shows the time-height cross-sections of ice concentration for turbulence levels set to 25% (left panel) and 75% (right panel) of the baseline, both with ice formation activated. In both cases, ice particles form within a relatively narrow altitude range around 82–86 km, reflecting the cold temperature regime required for noctilucent cloud (NLC) development. However, the lower-turbulence case (25%) exhibits a more confined layer of high ice concentration, whereas the 75% turbulence scenario displays a broader vertical extent. This suggests that enhanced turbulent mixing distributes ice particles over a wider altitude range, albeit with slightly lower peak concentrations in certain regions.

Figure 7 depicts the corresponding vertically averaged profiles of ice concentration for these same turbulence levels. The two profiles illustrate distinct maxima in ice concentration at approximately 82–84 km, consistent with the mesopause region where temperatures drop below the threshold required for water vapor condensation. Notably, the 25% turbulence run (blue line) exhibits a pronounced peak near 82 km, indicating a more localized region of ice growth, whereas the 75% run (orange line) shows a broader, slightly elevated peak that extends over a few kilometers in altitude. Together, these results emphasize the role of turbulence intensity in modulating not only the magnitude of ice formation but also its vertical distribution, a key factor in determining overall NLC brightness and spatial extent.

4.3. Additional Results

In addition to the water vapor and ice concentration fields presented earlier, Figures 9 and 8 illustrate further aspects of the model's output that were explored during initial investigations. Figure 9 shows the temporal evolution of meteoric dust particles, serving as potential condensation nuclei for noctilucent cloud (NLC) formation. Under lower turbulence (25%),

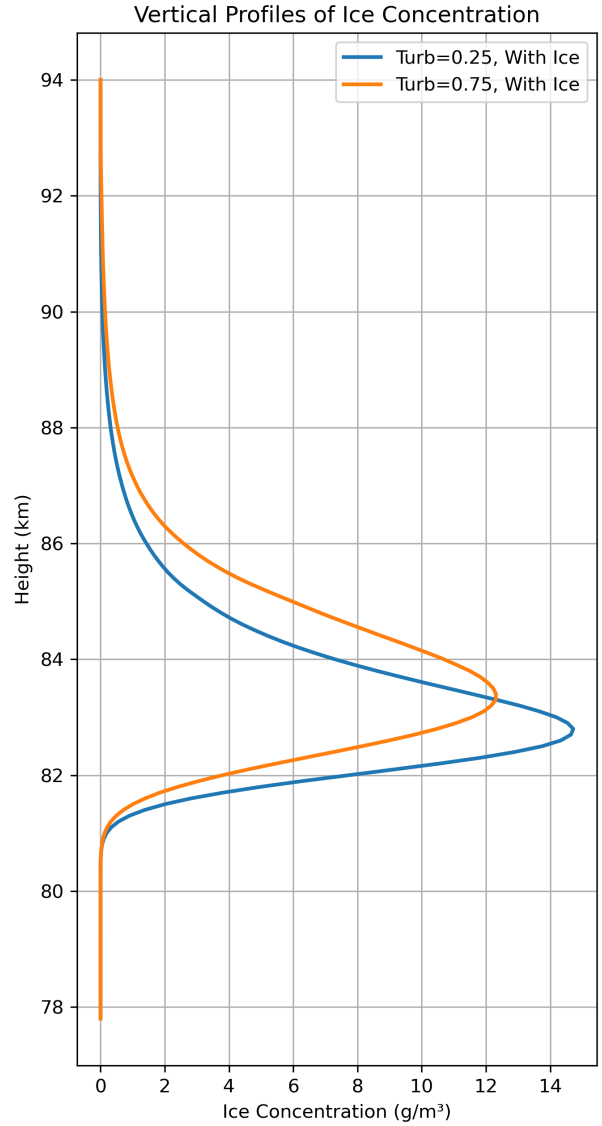


Figure 7: Vertically averaged profiles of ice concentration for the 25% (blue) and 75% (orange) turbulence cases. The pronounced peak between 82–84 km aligns with the cold mesopause region where noctilucent clouds typically form.

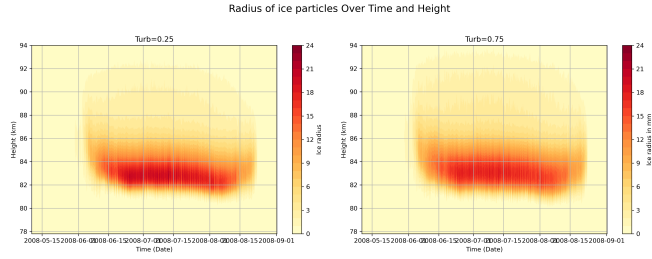


Figure 8: **Number of dust particles over time and height** for two turbulence cases (25% and 75% of baseline). The color scale indicates the number of dust particles per cubic centimeter in each time-height bin.

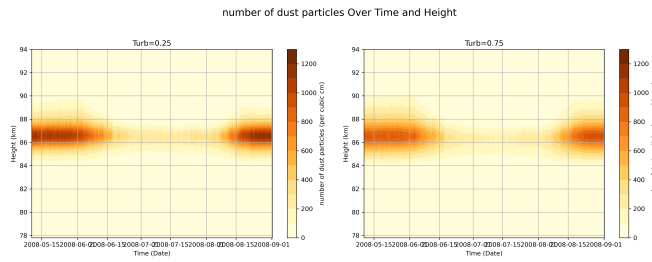


Figure 9: **Radius of ice particles (r_{ice}) over time and height** for turbulence intensities of 25% and 75% of baseline. Warmer colors correspond to larger particle radii, indicating more substantial growth under specific conditions.

dust appears more confined to a narrow altitude band, whereas higher turbulence (75%) disperses dust particles more broadly, potentially influencing where ice nucleation can occur.

Figure 8 presents the radius of ice particles (r_{ice}) as a function of time and height. Here, regions with larger radii indicate stronger or more prolonged ice growth, which may be facilitated by the presence of dust particles in supersaturated layers. The broader vertical extent of significant ice growth in the higher-turbulence case suggests that turbulence enhances mixing of both water vapor and dust, allowing for more widespread conditions favorable to particle growth.

4.3.1. Future Analysis and Additional Parameters

While the current figures highlight only a subset of the parameters examined, future analyses will incorporate additional diagnostics to provide a more comprehensive understanding of NLC formation under different turbulence regimes. Planned investigations include:

- **Ice Density ($dens_{ice}$):** Evaluating the mass density of ice within each grid cell to complement the radius-based analysis, offering insights into the total ice mass budget.
- **Particle Trajectory Tracking:** Following individual or representative groups of particles through time to ascertain their vertical and latitudinal movements under varying turbulence levels, thereby elucidating key transport mechanisms.
- **Latitudinal Extent of Ice Formation:** Determining the lowest latitude at which ice particles are detected under different turbulence and vertical wind conditions. This

will help clarify how far NLCs might extend toward mid-latitudes, potentially offering indicators of broader atmospheric changes.

These additional parameters, alongside continued refinement of the turbulence and vertical wind experiments, will further illuminate the multifaceted processes driving noctilucent cloud variability. As these analyses progress, more detailed results will be presented, contributing to a holistic view of NLC dynamics and their potential role as indicators of mesospheric and climate variability.

4.4. Outlook: Vertical Wind Experiments and Concluding Remarks

While the preliminary results presented herein focus on the role of turbulence in shaping noctilucent cloud (NLC) properties, a parallel set of experiments will systematically examine the influence of vertical wind variations. Similar to the turbulence sensitivity tests, the vertical wind will be scaled by factors below and above climatological baseline values, thereby capturing a range of possible scenarios from weak updrafts to more vigorous ascent. These controlled variations in vertical wind speed are crucial for understanding how adiabatic cooling and dynamical transport impact ice particle nucleation, growth, and eventual sublimation. Upward vertical winds may prolong particle residence in colder layers, fostering larger ice particles and more pronounced cloud brightness, whereas stronger downward winds could rapidly deplete clouds by transporting particles into warmer regions.

The integration of these vertical wind experiments with the turbulence simulations will allow us to disentangle the individual and combined effects of these two drivers. By comparing the spatiotemporal evolution of NLCs under different wind conditions, we aim to identify thresholds or synergy between turbulence-driven mixing and wind-driven vertical transport. This approach will also facilitate direct comparison against observational data—such as lidar and radar profiles—which often capture concurrent variability in both turbulence and vertical wind patterns.

4.5. Conclusion

In this report, we have outlined the scientific objectives and methodological framework for investigating the impacts of turbulence and vertical winds on noctilucent cloud formation using the Mesospheric Ice Microphysics And tranSPort (MIMAS) model. We presented preliminary findings illustrating how varying turbulence levels can significantly modify the distribution of water vapor, dust particles, and ice in the upper mesosphere. While these initial results are not definitive, they provide valuable insights into the microphysical processes and dynamical interactions shaping NLCs.

Building upon these turbulence-focused simulations, we will extend our analyses to encompass a range of vertical wind scenarios, systematically probing how updrafts and downdrafts influence ice particle nucleation, growth, and sublimation. These combined experiments promise to offer a comprehensive view of the mesospheric environment, elucidating how turbulence

and vertical wind, together with temperature and water vapor dynamics, govern the spatial and temporal variability of NLCs. Ultimately, this research will not only enhance our fundamental understanding of mesospheric processes but also support the development of more accurate models, guiding future observational campaigns and informing our broader knowledge of atmospheric and climate systems.

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